



Types of Computational Problems

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SUBJECT

Problem Solving Techniqu

SEMESTER

Semester 1

// Notes Summary

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title: "Types of Computational Problems" semester:
subject: "Problem Solving Techniques" subjectSlug:
"problemsolvingtechniques" unit: "Unit I: Problem
Solving Computational Thinking" unitSlug:
"unit1problemsolvingsteps" order: description:
"Classification of computational problems into
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Topics Covered

Intermediate

Difficulty Level



Why this topic exists

Not all computational problems share the same structure or operational goals.

Classifying a problem into its fundamental category dictates the algorithmic strategy required—determining whether a solution exists requires a different approach than finding the optimal solution or computing the total number of valid solutions.

Categorizing problems establishes a systematic framework for evaluating computational complexity and selecting appropriate data structures.

Core Categories of Computational Problems

♦ [DEFINITION] 1. Decision Problem

A problem that requires a binary output—strictly **Yes/No** or **True/False**—based on whether a given input satisfies a specific condition.

- **Key Trait:** Evaluates a single logical property without returning explicit data objects or optimal configurations. - **Examples:** - "Is an integer n prime?" - "Does a given graph contain a cycle?" - "Is a given array sorted in non-decreasing order?"

♦ [DEFINITION] 2. Searching Problem

A problem that requires locating a target element or determining its position/index within a structured dataset.

- **Key Trait:** Yields a location, reference, or explicit instance of an existing data item within a collection. - **Examples:** - "Locate the index of target value k in array A ." - "Retrieve a specific user record from a database using its Primary Key." - "Find a node with value x in a binary search tree."



✦ [DEFINITION] 3. Optimization Problem

A problem where multiple valid solutions exist, and the objective is to find the solution that minimizes or maximizes a specific objective function.

- **Key Trait:** Evaluates candidate solutions against a quantitative metric to identify the single optimal option. - **Examples:** - "Find the shortest path between two vertices in a weighted graph." - "Determine the maximum total value of items that fit into a knapsack of capacity W ." - "Find the minimum number of coins required to make change for a given amount."

✦ [DEFINITION] 4. Counting Problem

A problem that requires determining the total number of valid solutions or configurations that satisfy a given set of constraints, without enumerating each solution explicitly.

- **Key Trait:** Yields a single non-negative integer representing the total count of valid instances. - **Examples:** - "Calculate the total number of distinct paths from top-left to bottom-right in an $m \times n$ grid." - "Find the number of distinct ways to make change for ₹10 using given coin denominations." - "Determine the number of valid spanning trees in a connected graph."

Comparative Overview

Problem Type	Target Question	Expected Output Type	Primary Objective
Decision	Is condition met?	Binary (<code>True</code> / <code>False</code>)	Verification
Searching	Where is target located?	Index / Reference / Pointer	Data Retrieval
Optimization	What is best solution?	Optimal Metric / Config	Cost Minimization / Yield Maximization
Counting	How many valid solutions exist?	Non-negative Integer ($c \in \mathbb{N}_0$)	Combinatorial Enumeration

Problem Classification Guide

To classify a problem statement quickly, analyze the leading phrasing and functional requirement:



- **Decision Indicators:** *"Is...", "Does...", "Can...", "Check if..."*
- **Searching Indicators:** *"Locate...", "Find the position of...", "Search for..."*
- **Optimization Indicators:** *"Find the shortest...", "Maximize...", "Minimize...", "Find the optimal..."*
- **Counting Indicators:** *"Count...", "How many ways...", "Find the total number of..."*

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